

IrDA Link Negotiation: Optimizing Minimum Turnaround Time

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Abstract—IrDA has become a popular standard for wireless point-to-point communications, and is increasingly being used for high-bandwidth applications. In order to optimize performance at high data rates, an accurate understanding of the physical and protocol-related components that influence IrDA throughput is necessary. During link establishment, link attributes are negotiated between IrDA devices, including the minimum amount of time that must elapse during link turnaround before a sending device is ready to receive data. This parameter is called *minimum turnaround time*. Our research expands upon a model for IrDA throughput and specifically analyzes device latencies and the granularity of IrDA's minimum turnaround time parameter and their effect on throughput. We demonstrate that the limited options for minimum turnaround time dictated in current IrLAP implementations can significantly degrade throughput, depending on a device's capabilities. We also identify two negotiation strategies which devices may employ to select an optimal minimum turnaround time.

I. INTRODUCTION

Used on over 300 million devices worldwide, the IrDA protocol has become a popular standard for wireless connectivity. A wide variety of products now support this low-cost method of wireless data transmission, including notebook computers, cellular phones, digital cameras and PDAs. As the IrDA market has grown and matured, the range of its possible applications has multiplied, and there has been an increasing demand for higher levels of data throughput and performance. Factors that once had a minor or insignificant impact on performance at lower transmission rates have become increasingly relevant at higher data rates. An accurate understanding of the physical and protocol-related components that influence IrDA throughput is necessary to optimize performance.

IrDA is a short-range point-to-point protocol for wireless communication that relies on pulses of infrared light to transmit data. Residing just below the spectrum of visible light, the infrared medium is inherently noisy and is easily interfered with by other light sources. Because IrDA is half-duplex, each device must wait its turn while the other transmits to avoid interfering with the light from the signal it is receiving. This constraint introduces some complexities into the IrDA specification, among which is the necessity of periodically turning the link around in order to achieve bi-directional communication.

During link establishment, certain link characteristics are negotiated between devices, including the minimum amount of

time that must elapse during link turnaround before a sending device is ready to receive data. This parameter is called the *minimum turnaround time*. Our research expands upon a model for IrDA throughput and specifically analyzes device latencies and the granularity of IrDA's minimum turnaround time parameter and their effect on throughput. We demonstrate that the limited options for minimum turnaround time dictated in current IrLAP implementations can significantly degrade throughput, depending on a device's capabilities. We also identify two negotiation strategies devices may employ for choosing an optimal minimum turnaround time.

II. BACKGROUND

IrDA is a multi-layered protocol consisting of three mandatory components: the IrDA physical layer (IrPHY), the IrDA data link layer (IrLAP), and the IrDA link management layer (IrLMP). Additional higher layer protocols may also be supported in IrDA devices, including: TinyTP, an application-level flow control protocol; IrCOMM, a serial and parallel port link emulator that allows applications designed to work with serial and parallel ports to operate unchanged over an IR link; IrLAN, which allows access to a LAN over an IrDA link; and IrOBEX, which provides support for basic data object exchange. See Figure 1 for a diagram of the IrDA protocol stack. IrDA throughput can be modeled accurately by considering parameters that are negotiated in the first two layers of the protocol stack. However, it is also important to optimize higher layer settings for parameters negotiated in the IrLAP layer, as has been demonstrated by Deccio et al in their analysis of the IrOBEX protocol [1].

A. IrPHY Layer

The physical layer consists of the optical, optoelectronic and electronic components necessary to transmit data across the IrDA link. It also provides a *frame check sequence* (FCS) for error detection, and mechanisms for the proper framing and division of data going over the link. Each successive version of the IrPHY specification supports higher data rates: the Serial IR (SIR) specification operates at a maximum speed of 115.2 Kbps; FIR supports data rates of up to 4 Mbps, and VFIR supports a single data rate of 16 Mbps.

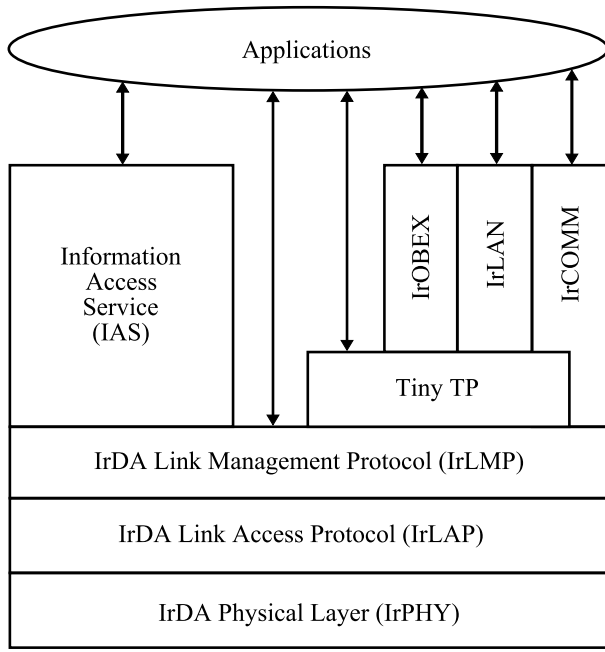


Fig. 1. IrDA Stack

B. IrLAP Layer

IrLAP provides device discovery, connection establishment, and data exchange services to upper layers. The device that initiates the discovery process assumes the role of the *primary* device, and all other devices involved become *secondary* devices. Although the IrLAP discovery process was designed to accommodate more than one secondary device, protocols for point-to-multipoint data exchange have not yet been specified. Hence, our research is confined to conventional point-to-point IR communications. In addition to simple device discovery, IrLAP allows each device to provide certain “hints” about the services that it supports, which facilitates service discovery at the application level. The discovery and connection establishment processes take place at a predefined data rate of 9600 bps.

After device discovery, connection establishment may occur. During connection establishment, the primary device advertises its capabilities, including supported data rates and device latency. The secondary device responds with its supported parameter options, and the devices then choose options in harmony with the capabilities of both devices. Some parameters must be agreed upon (such as data rate), while others need only be respected. The parameters negotiated during connection establishment directly affect throughput, and are therefore essential components of our IrDA throughput model.

Among the parameters negotiated during connection establishment is minimum turnaround time. This parameter specifies the time that each device requires after sending data before it is ready to receive the next transmission. Current IrLAP implementations support only the following eight minimum turnaround time values: 10, 5, 1, 0.5, 0.1, 0.05, 0.01, or

TABLE I
MODEL PARAMETER DEFINITIONS

Symbol	Parameter Description	Unit
p_b	Link bit error rate	-
p_f	Frame error probability	-
p_{te}	Turnaround error probability	-
p_{ff}	First frame error probability	-
n_{cf}	Correctly transmitted frames	-
n_{te}	Number of turnaround errors possible	-
l	I-frame message data length	bits
l'	S-frame length	bits
t_i	Transmission time of an I-frame	ms
$t_{I_{max}}$	Transmission time of a 16Kbit I-frame	ms
t_s	Transmission time of an S-frame	ms
t_{ld}	Link disconnect threshold	ms
t_{ack}	Acknowledgment time	ms
t_{max}	Maximum turnaround time	ms
$t_{F_{out}}$	F-timer time-out period	ms
$\bar{t}$	Average window transmission time	ms
$D(t)$	Device latency distribution	-
N	Window size	-

0 ms. The discrepancy between a device’s actual latency and the specification’s options for minimum turnaround time can impact throughput. It is therefore important to consider these factors when negotiating minimum turnaround times. Our throughput model provides a simple method for choosing optimal minimum turnaround time values.

III. RELATED WORK

Mathematical models of IrLAP throughput have been developed by others. In [2], Robertson et al developed a mathematical model for the IrDA protocol and implemented it in software. They were able to analyze and simulate IrDA data transfers using different values for negotiation parameters. Boucouvalas and others in [3], [4], [5], [6], and [7] developed an analytical mathematical model for IrLAP throughput based on certain key negotiation parameters. The forward-looking model in [3] identifies optimal values for window size and frame size for future IrLAP specifications with data rates up to 100 Mbps. In this model a primary device sends full windows to a secondary, which responds only with a supervisory frame, acknowledging correctly received frames. They concluded, not surprisingly, that the best throughput is achieved with a large window size and small minimum turnaround values. Other papers such as [8] provide a summary and future predictions for the IrDA protocol itself.

IV. IRLAP MODEL

We extend the work of Boucouvalas by considering the possibility that the negotiated minimum turnaround time is either not long enough or too long. See Table I for model parameter definitions.

We assume that the actual time necessary for preparation to receive can be described probabilistically by a latency distribution. This preparation time consists of hardware and software components. Relevant hardware components, including the infrared transmitter and receiver, reside in the same chamber.

During transmission, the chamber is flooded with light, rendering the receiver incapable of detecting signals from other devices until the energy has dissipated. This dissipation time varies yielding software latencies stemming from interrupt delays and processing overhead. The variation in these factors can be characterized by a probabilistic distribution, which we term the *latency distribution*. The latency distribution lies between a minimum time, below which the device is never ready, and a maximum time, above which the device will always be ready. A sample latency distribution can be seen in Figure 2.

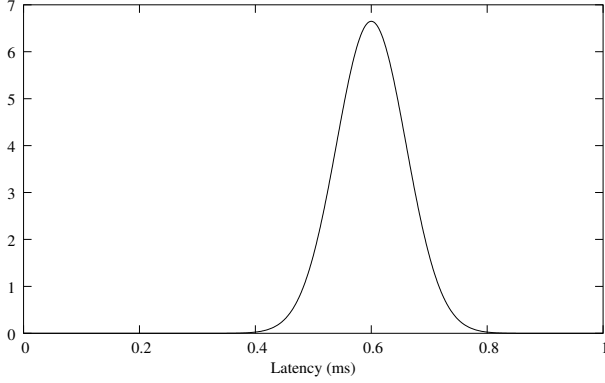


Fig. 2. Device latency distribution: $Normal(\mu = 0.6, \sigma = 0.06)$

If a device negotiates a minimum turnaround time below its maximum latency, there is a possibility that it will not be prepared to receive when another device begins to transmit, resulting in what we call a *turnaround error*. The probability of a turnaround error p_{te} is determined by the negotiated minimum turnaround time and the device's latency distribution.

$$p_{te}(t) = \int_t^{t_{max}} D(t)dt \quad (1)$$

The time period after a device receives and before it responds may also be described probabilistically. However, our model presumes that devices respond immediately.

Throughput is the average number of correctly transmitted frames in a window divided by the average window transmission time. The window transmission time is the average time from the start of a window transmission to the start of the next window transmission. In [3] Boucouvalas and Vitsas derive an equation for the average window transmission time that considers the bit error rate:

$$\bar{t} = Nt_i + p_f(t_{Fout} + t_s) + t_{ack} \quad (2)$$

We now incorporate a time penalty term for turnaround errors. When a turnaround error occurs because the primary device is not ready, the time penalty incurred is $t_{Fout} + t_s$. There is no time penalty for turnaround errors when the secondary device is not ready. The link disconnect threshold, t_{ld} , limits the number of consecutive turnaround errors that

may occur within the same transmission window before link disconnection. This limit is

$$n_{te} = \frac{t_{ld}}{t_{Fout} + t_s} \quad (3)$$

Our new equation for the average window transmission time is

$$\bar{t} = Nt_i + p_f(t_{Fout} + t_s) + \sum_{k=1}^{n_{te}} p_{te}^k(t_{Fout} + t_s) + t_{ack} \quad (4)$$

In [3] Boucouvalas and Vitsas derive the following equation for the average number of correctly transmitted frames in a transmission window.

$$n_{cf} = N(1 - p_f)^N + \sum_{w=1}^{N-1} w(1 - p_f)^w p_f \quad (5)$$

If we consider turnaround errors, the first frame in a transmission window may be lost if the secondary device is not ready. Losing the first frames requires retransmission of the entire window, thus reducing the average number of correctly transmitted frames. Our new equation for the average number of correctly transmitted frames in a window is

$$n_{cf} = N(1 - p_{ff})(1 - p_f)^{N-1} + \sum_{w=1}^{N-1} w(1 - p_{ff})(1 - p_f)^{w-1} p_f \quad (6)$$

Figure 3 shows the incremental development of our model with different transmission scenarios and error possibilities. Each box represents a single frame. The three characters in each frame indicate respectively the frame type (information i , or supervisory s), the sequence number, and the poll / final bit. Frames that are crossed out represent a lost frame due to either a bit error or turnaround error. See Table I for a description of the time variables. Of particular interest is Part d, which considers turnaround errors occurring in both the primary and secondary devices.

V. THROUGHPUT ANALYSIS

The following results are based on our model given primary and secondary devices with a 1.0 ms advertised minimum turnaround time and a normal latency distribution (Figure 2) with mean 0.6 and standard deviation 0.06. The negotiated minimum turnaround time of the secondary device is fixed at 1.0 ms.

A. Throughput Losses Due to IrLAP Granularity

Throughput losses due to IrLAP's limited choices for minimum turnaround time can be significant, depending on a device's latency distribution. Figure 4 shows the difference in throughput between continuously varying minimum turnaround times and IrLAP-allowed minimum turnaround times. For example, if a device's maximum latency is 1.1 ms, and it negotiates a minimum turnaround time of 5.0 ms, the resulting throughput loss will be about 450 kbps. With a negotiated minimum turnaround time of zero there is close to a 100% chance of a turnaround failure (See Equation

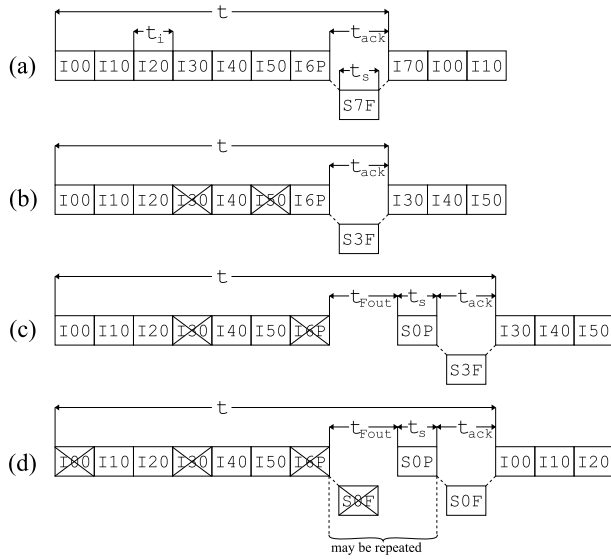


Fig. 3. (a) Error free transmission of a window. (b) Retransmitted frames due to bit errors in frames with sequence numbers 3 and 5. (c) Retransmitted frames and F-timer delay due to bit errors in frames with sequence numbers 3 and 6. (d) F-timer delay due to turnaround error on acknowledgment from secondary and then entire window retransmitted due to turnaround error in frame with sequence number 0. Adapted from [3]

1), and throughput is nearly zero. The negative differences in throughput can be explained with the following example. Low negotiated minimum turnaround times (far below the average latency) result in a high probability of turnaround failure. Thus, negotiating at the next highest IrLAP-allowed minimum turnaround time gives greater throughput in this case. However, differences in throughput become positive at higher negotiated minimum turnaround times because the probability of turnaround failure is lower. Additionally, the delays from higher IrLAP-allowed values decrease throughput. It is also interesting to note that the differences in throughput are greater at higher data rates.

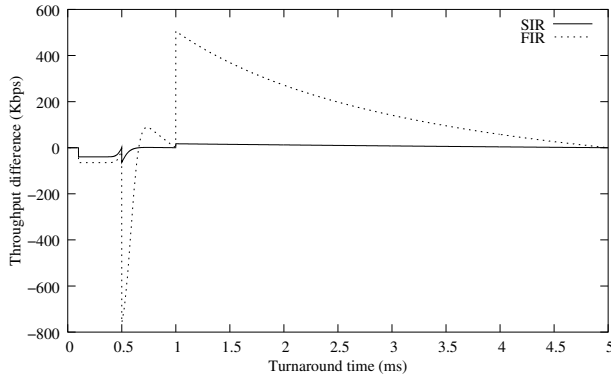


Fig. 4. Difference in throughput between continuously-varying minimum turnaround times and IrLAP-allowed minimum turnaround times

B. Minimum Turnaround Time Negotiation Strategies

IrLAP's limited minimum turnaround time possibilities suggest two negotiation strategies. Given a maximum device latency value and the IrLAP-allowed minimum turnaround times, a device can negotiate the next highest or next lowest allowed value.

1) *Conservative Negotiation*: Negotiating a minimum turnaround time above the maximum device latency will result in zero turnaround failures. We call this strategy *conservative negotiation* (See Figure 5). Conservative negotiation is appropriate when a device's maximum latency is slightly below an IrLAP minimum turnaround time possibility. This is the strategy employed by most IrDA devices today.

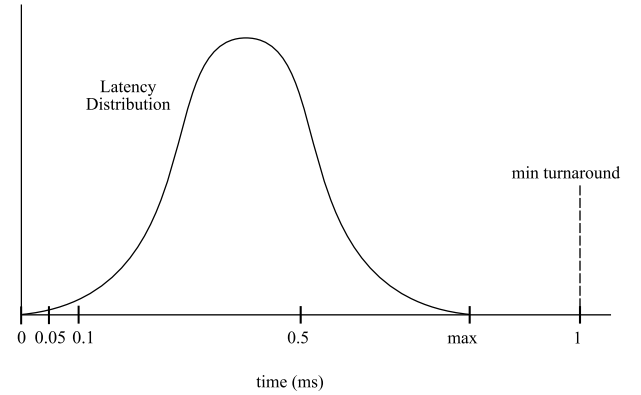


Fig. 5. Conservative Negotiation: The vertical dashed line represents the advertised minimum turnaround time.

2) *Aggressive Negotiation*: While conservative negotiation eliminates turnaround errors, higher turnaround times reduce throughput. Negotiating an IrLAP-allowed minimum turnaround time below the maximum latency may result in turnaround errors. However, shorter turnaround times increase throughput. We call this strategy *aggressive negotiation* (See Figure 6). Aggressive negotiation is appropriate when a device's maximum latency is slightly above an IrLAP-allowed minimum turnaround time possibility. The area of the shaded region in Figure 6 represents the probability of a turnaround error.

For the device latency distribution in Figure 2, conservative negotiation yields the highest throughput. The maximum device latency is about 0.78 ms. Conservative negotiation would choose 1.0 ms, the next highest IrLAP-allowed minimum turnaround time. Aggressive negotiation would choose 0.5 ms, the next lowest IrLAP-allowed minimum turnaround time. Figure 7 shows that a 1.0 ms minimum turnaround time yields higher throughput than 0.5 ms.

For the device latency distribution in Figure 8, aggressive negotiation yields the highest throughput. The maximum device latency is about 0.58 ms. Aggressive negotiation would choose 0.5 ms, the next lowest IrLAP-allowed minimum turnaround time. Conservative negotiation would choose 1.0 ms, the next highest IrLAP-allowed minimum turnaround time.

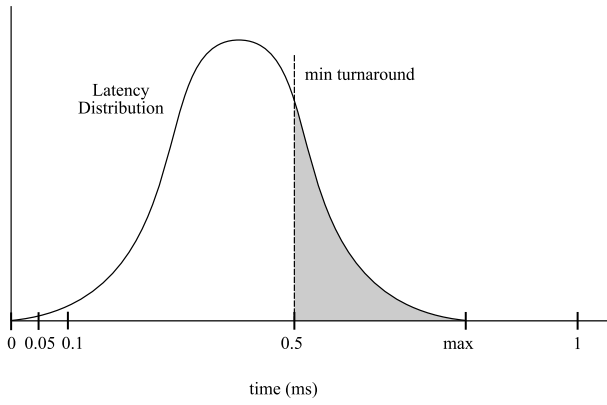


Fig. 6. Aggressive Negotiation: The vertical dashed line represents the advertised minimum turnaround time.

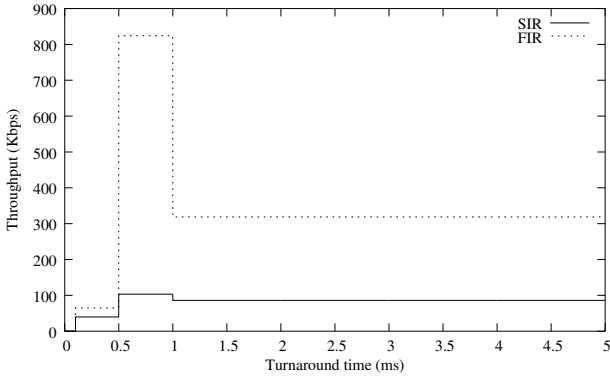


Fig. 7. Throughput for IrLAP-allowed minimum turnaround times for device latency distribution in Figure 2

Figure 9 shows that a 0.5 ms minimum turnaround time yields higher throughput than 1.0 ms.

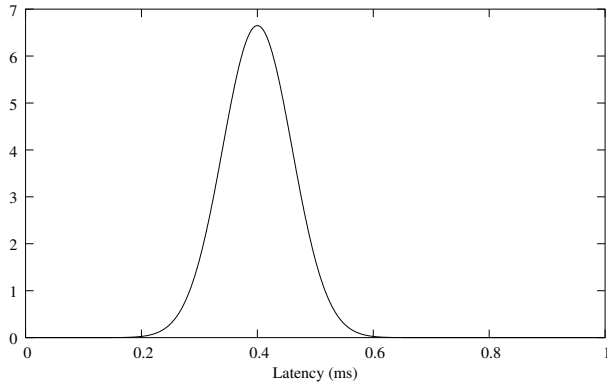


Fig. 8. Device latency distribution: $Normal(\mu = 0.4, \sigma = 0.06)$

VI. OPTIMAL MINIMUM TURNAROUND TIME

A new addendum to the IrLAP 1.1 specification [9] adds support for a new data rate, VFIR (16 Mbps) and scalar representation of negotiation parameters. A scalar representation of the minimum turnaround time parameter can be used in

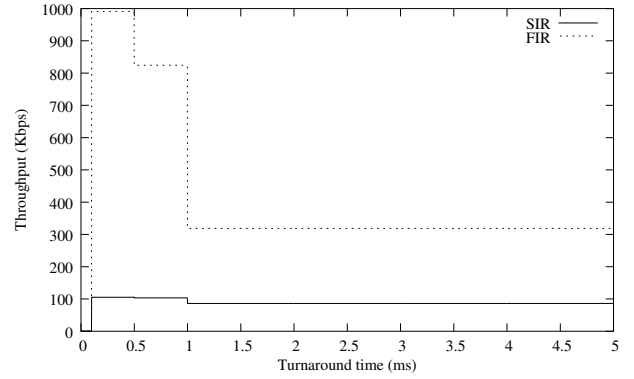


Fig. 9. Throughput for IrLAP-allowed minimum turnaround times for device latency distribution in Figure 8

SIR and FIR, however in VFIR the minimum turnaround time is fixed at $100 \mu s$. Thus, in SIR and FIR, rather than being limited to eight possible minimum turnaround time values, a device can select from 65,536 possible values.

Scalar representation makes it possible to choose the optimum minimum turnaround time for a given device's hardware and software capabilities. Figure 10 shows throughput using the scalar representation for minimum turnaround time for the device latency distribution in Figure 2. The optimal minimum turnaround time of 0.733 ms can be negotiated rather than being limited to 0.5 ms or 1 ms.

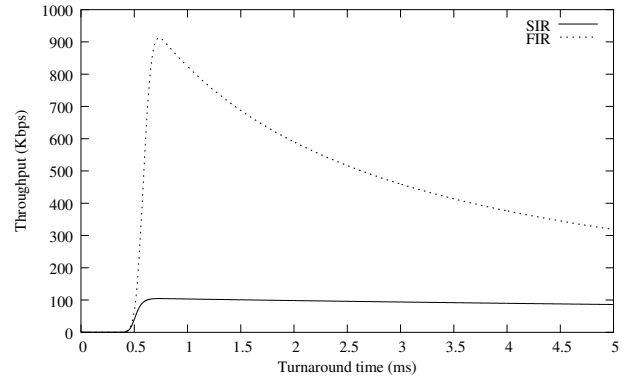


Fig. 10. Throughput for continuously-varying minimum turnaround times

VII. CONCLUSIONS

We have shown that minimum turnaround time and a device's latency distribution are significant factors in IrDA throughput. As the minimum turnaround time decreases, the throughput increases until the minimum turnaround time reaches a point where turnaround errors overpower throughput gains. These effects are even more pronounced at higher data rates (such as the 4 Mbps used in FIR).

An optimum value for a device's minimum turnaround time can be calculated using the mathematical models that we have presented. This optimum value achieves a balance of low minimum turnaround time with few turnaround errors and

largely depends on a given device's latency distribution. Our model identifies optimal minimum turnaround times for devices regardless of the representation used (scalar or granular).

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